

Vueports Solutions — Local Development Guide

Version: 1.0 | Last Updated: 2026-05-27 | Target: Windows, macOS, Linux

Table of Contents

1. Quick Start (5 Minutes)
2. Option A: XAMPP (Windows/Linux/macOS)
3. Option B: Laragon (Windows)
4. Option C: Docker (All Platforms)
5. Option D: Manual LAMP/WAMP/MAMP
6. Database Setup
7. Environment Configuration
8. First Run Checklist
9. Development Workflow
10. Common Local Issues
11. VS Code Setup
12. Testing PayFast Locally

1. Quick Start (5 Minutes)

Prerequisites Check

```
# Check if you already have PHP
php -v

# Check if you already have MySQL
mysql -V

# Check if you already have Apache
apachectl -v
```

If any are missing, pick an option below. **Recommended for beginners:** Option A (XAMPP) or Option B (Laragon).

2. Option A: XAMPP

Best for: Beginners, all platforms, quick setup

2.1 Download & Install

1. Go to <https://www.apachefriends.org>
2. Download **XAMPP 8.2.x** (PHP 8.2, MySQL 8.0)
3. Run installer
 - **Windows:** Run as Administrator
 - **macOS:** Drag to Applications

- **Linux:** `chmod +x xampp-linux-x64-8.2.x-installer.run && sudo ./xampp-linux-x64-8.2.x-installer.run`

2.2 Start Services

```
# Windows: Open XAMPP Control Panel -> Start Apache + MySQL
# macOS: Open XAMPP Manager -> Start Services
# Linux: sudo /opt/lampp/lampp start
```

2.3 Project Setup

```
# Windows
C:\xampp\htdocs\vueports\

# macOS
/Applications/XAMPP/htdocs/vueports/

# Linux
/opt/lampp/htdocs/vueports/
```

Copy all project files into the vueports folder.

2.4 PHP Configuration (php.ini)

```
# Windows: C:\xampp\php\php.ini
# macOS: /Applications/XAMPP/etc/php.ini
# Linux: /opt/lampp/etc/php.ini
```

Edit these values:

```
upload_max_filesize = 10M
post_max_size = 10M
max_execution_time = 300
memory_limit = 256M
max_input_vars = 3000

; Enable these extensions (remove ; if present)
extension=openssl
extension=pdo_mysql
extension=mbstring
extension=curl
extension=gd
extension=zip
extension=bcmath
extension=intl

; OPcache for local dev (optional)
zend_extension=opcache
opcache.enable = 1
opcache.memory_consumption = 64
opcache.validate_timestamps = 1 ; Important: 1 for dev, 0 for production
```

Restart Apache after changes.

2.5 MySQL Access

```
# Windows: http://localhost/phpmyadmin
# Default login: root / (no password)

# Or command line
# Windows: C:\xampp\mysql\bin\mysql.exe -u root
# macOS/Linux: /opt/lampp/bin/mysql -u root
```

3. Option B: Laragon

Best for: Windows developers, modern stack, auto-SSL, quick virtual hosts

3.1 Download & Install

1. Go to <https://laragon.org>
2. Download **Laragon Full** (~150 MB)
3. Install to C:\laragon
4. Laragon auto-starts — click **Start All**

3.2 Project Setup

```
# Laragon uses auto-virtual-hosts
# Just place project in C:\laragon\www\vueports\
# Access at: http://vueports.test/ (auto-generated!)
```

3.3 Enable Required Extensions

```
# Laragon Menu -> PHP -> Extensions
# Check: openssl, pdo_mysql, mbstring, curl, gd, zip, bcmath, intl
```

3.4 Database

```
# Laragon Menu -> MySQL -> HeidiSQL (GUI)
# Or: Laragon Menu -> Terminal -> mysql -u root
```

3.5 SSL (Auto)

```
# Laragon auto-generates SSL certificates
# Access https://vueports.test/ automatically
```

4. Option C: Docker

Best for: Team consistency, matching production, advanced users

4.1 Install Docker

- Docker Desktop for Windows/macOS: <https://www.docker.com/products/docker-desktop>
- Linux: `sudo apt install docker.io docker-compose`

4.2 Docker Compose File

Create `docker-compose.yml` in project root:

```
version: '3.8'

services:
  app:
    image: php:8.2-apache
    container_name: vueports-app
    ports:
      - "8080:80"
    volumes:
      - ../var/www/html
      - ./docker/php.ini:/usr/local/etc/php/php.ini
    environment:
      - APACHE_DOCUMENT_ROOT=/var/www/html
    command: >
      bash -c "
        a2enmod rewrite headers deflate expires ssl &&
        docker-php-ext-install pdo pdo_mysql mysqli gd zip bcmath intl opcache &&
        apache2-foreground
      "
    depends_on:
      - db
      - redis

  db:
    image: mysql:8.0
    container_name: vueports-db
    ports:
      - "3306:3306"
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: vueports_db
      MYSQL_USER: vueports_user
      MYSQL_PASSWORD: vueports_pass
    volumes:
      - mysql_data:/var/lib/mysql
      - ./database.sql:/docker-entrypoint-initdb.d/01-schema.sql

  redis:
    image: redis:7-alpine
```

```
container_name: vueports-redis
ports:
  - "6379:6379"

phpmyadmin:
  image: phpmyadmin/phpmyadmin
  container_name: vueports-pma
  ports:
    - "8081:80"
  environment:
    PMA_HOST: db
    PMA_PORT: 3306

volumes:
  mysql_data:
```

4.3 Start Containers

```
cd /path/to/vueports-project
docker-compose up -d

# View logs
docker-compose logs -f app

# Stop
docker-compose down

# Full reset (wipes database!)
docker-compose down -v && docker-compose up -d
```

4.4 Access Points

Service	URL
Website	http://localhost:8080
phpMyAdmin	http://localhost:8081
MySQL	localhost:3306
Redis	localhost:6379

4.5 Run Commands Inside Container

```
# PHP CLI
docker-compose exec app php -v

# MySQL
docker-compose exec db mysql -u vueports_user -p vueports_db

# Composer
docker-compose exec app composer install
```

```
# Run cron scripts
docker-compose exec app php cron/daily-backup.php
```

5. Option D: Manual

5.1 Windows (WAMP)

1. Download WampServer: <https://www.wampserver.com/>
2. Install PHP 8.2, MySQL 8.0, Apache 2.4
3. Place project in C:\wamp64\www\vueports\
4. Access at <http://localhost/vueports/>

5.2 macOS (MAMP)

1. Download MAMP: <https://www.mamp.info/>
2. Place project in /Applications/MAMP/htdocs/vueports/
3. Access at <http://localhost:8888/vueports/>

5.3 Linux (Manual LAMP)

```
# Ubuntu/Debian
sudo apt update
sudo apt install apache2 mysql-server php8.2 libapache2-mod-php8.2 php8.2-mysql php8.2-curl php8.2-gd
↪ php8.2-mbstring php8.2-xml php8.2-zip php8.2-bcmath php8.2-intl php8.2-openssl

# Enable Apache rewrite
sudo a2enmod rewrite
sudo systemctl restart apache2

# Project location
sudo mkdir -p /var/www/vueports
sudo chown $USER:$USER /var/www/vueports

# Copy project files here
# Access at http://localhost/
```

6. Database Setup

6.1 Create Database

```
-- Via phpMyAdmin or MySQL CLI
CREATE DATABASE vueports_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'vueports_user'@'localhost' IDENTIFIED BY 'localpass';
GRANT ALL PRIVILEGES ON vueports_db.* TO 'vueports_user'@'localhost';
FLUSH PRIVILEGES;
```

6.2 Import Schema

```
# Command line
```

```
mysql -u vueports_user -p vueports_db < database.sql

# Or via phpMyAdmin -> Import -> Choose file
```

6.3 Verify Tables

```
USE vueports_db;
SHOW TABLES;
-- Should show 26 tables
```

7. Environment Configuration

7.1 Create .env

```
cp .env.example .env
```

7.2 Local .env Template

```
# =====
# LOCAL DEVELOPMENT ENVIRONMENT
# =====

APP_NAME="Vueports Solutions"
APP_URL=http://localhost
APP_ENV=development
APP_DEBUG=true

# Database (match your setup)
DB_HOST=localhost
DB_NAME=vueports_db
DB_USER=vueports_user
DB_PASS=localpass

# SMTP (use Mailtrap for local dev)
SMTP_HOST=smtp.mailtrap.io
SMTP_USER=YOUR_MAILTRAP_USER
SMTP_PASS=YOUR_MAILTRAP_PASS
SMTP_PORT=2525
SMTP_FROM=test@vueports.local
SMTP_FROM_NAME="Vueports Dev"

# PayFast (SANDBOX ONLY for local!)
PAYFAST_MERCHANT_ID=10000100
PAYFAST_MERCHANT_KEY=46f0cd694581a
PAYFAST_PASSPHASE=
PAYFAST_SANDBOX=true
PAYFAST_URL=https://sandbox.payfast.co.za/eng/process
PAYFAST_ITN_URL=https://sandbox.payfast.co.za/eng/query/validate
```

```

# reCAPTCHA (use test keys for local)
RECAPTCHA_SITE_KEY=6LeIxAcTAAAAAJcZVRqyHh71UMIEGNQ_MXjiZKhI
RECAPTCHA_SECRET_KEY=6LeIxAcTAAAAAGG-vFI1TnRWxMZNFuojJ4WifJWe

# JWT (any random string for local)
JWT_SECRET=local_dev_secret_key_change_in_production
JWT_EXPIRY=3600

# Encryption
ENCRYPTION_KEY=local_dev_encryption_key

# Redis (optional)
REDIS_HOST=127.0.0.1
REDIS_PORT=6379
CACHE_DRIVER=file

# Session
SESSION_LIFETIME=120
SESSION_SECURE=false          # false for local HTTP
SESSION_HTTPONLY=true
SESSION_SAMESITE=Lax

# Rate Limiting
RATE_LIMIT_ENABLED=false      # disable for local testing

# Backup
BACKUP_S3_BUCKET=
AWS_ACCESS_KEY=
AWS_SECRET_KEY=

# Monitoring
HEALTH_CHECK_KEY=local
ERROR_REPORTING_EMAIL=

# WebSocket
WEBSOCKET_HOST=127.0.0.1
WEBSOCKET_PORT=8080
WEBSOCKET_SSL=false

# Zapier
ZAPIER_API_KEY=local_test_key

```

7.3 Mailtrap Setup (Recommended for Local Email Testing)

1. Sign up at <https://mailtrap.io>
2. Create inbox -> get SMTP credentials
3. Paste into .env SMTP_* fields
4. All emails are captured in Mailtrap inbox (never sent to real addresses)

8. First Run Checklist

Open browser and verify each page:

URL	Expected Result
http://localhost/	Homepage loads with logo, services, stats
http://localhost/about.php	About page with company info
http://localhost/services.php	3 service cards displayed
http://localhost/service-detail.php?slug=custom-software-web	Individual service page
http://localhost/calculator.php	Cost estimator works
http://localhost/consultation.php	Form submits to database
http://localhost/booking.php	Calendar picker works
http://localhost/portfolio.php	Project grid
http://localhost/contact.php	Contact form + map
http://localhost/pay.php?plan=starter	PayFast sandbox checkout
http://localhost/blog/	Blog listing (empty initially)
http://localhost/legal/privacy.php	Privacy policy

Admin Portal

URL	Login
http://localhost/admin/login.php	admin / admin123
http://localhost/admin/dashboard.php	Stats, messages, bookings
http://localhost/admin/analytics.php	Traffic charts

Client Portal

URL	Action
http://localhost/client/register.php	Create test account
http://localhost/client/login.php	Login with new account
http://localhost/client/dashboard.php	Project overview
http://localhost/client/projects.php	Project list

9. Development Workflow

9.1 File Structure for Development

```
vueports/
├── .env                <- Edit for local settings
├── .env.example        <- Template (don't edit)
├── config/
│   ├── database.php    <- Reads from .env
│   └── env.php          <- Environment loader
├── includes/
│   ├── functions.php    <- Core utilities
│   ├── header.php       <- Page header template
│   ├── footer.php       <- Page footer template
│   └── emails.php       <- Email functions
├── admin/              <- Admin portal pages
├── client/             <- Client portal pages
├── api/v1/             <- REST API endpoints
├── assets/
│   ├── css/            <- Edit styles here
│   └── js/              <- Edit scripts here
├── database.sql         <- Schema + seed data
└── docker-compose.yml   <- If using Docker
```

9.2 Making Changes

```
# 1. Edit files in your IDE
# 2. Refresh browser (no restart needed for PHP)
# 3. For CSS/JS changes: hard refresh (Ctrl+Shift+R)

# If using Docker:
docker-compose exec app php -l file.php  # Syntax check
```

9.3 Database Migrations (Manual)

```
# Export current schema
mysqldump -u vueports_user -p --no-data vueports_db > schema_update.sql

# Or add new table manually:
mysql -u vueports_user -p vueports_db
```

```
-- Example: Add new column
ALTER TABLE clients ADD COLUMN newsletter_opt_in TINYINT(1) DEFAULT 0;
```

9.4 Reset Everything

```
# Reset database (keeps schema, removes data)
mysql -u root -p -e "DROP DATABASE vueports_db; CREATE DATABASE vueports_db;"
mysql -u vueports_user -p vueports_db < database.sql

# Clear cache
```

```
rm -rf cache/ratelimit/*
rm -rf logs/*

# Clear uploads (test files)
rm -rf uploads/invoices/* uploads/projects/* uploads/avatars/*
```

10. Common Local Issues

Issue	Solution
“Connection refused”	MySQL not running -> Start XAMPP/Laragon MySQL service
“Access denied for user”	Wrong DB credentials in .env -> Check username/password
“Class PDO not found”	pdo_mysql extension not enabled -> Edit php.ini
** “Undefined function openssl_” ***	extension=openssl missing in php.ini
“404 on all pages except index”	mod_rewrite not enabled -> a2enmod rewrite (Linux) or check XAMPP Apache config
“Cannot write to logs/”	Permission denied -> chmod 775 logs/
“PayFast payment fails”	Using live keys locally -> Switch to sandbox keys in .env
“Emails not sending”	SMTP blocked locally -> Use Mailtrap credentials
“CSS not updating”	Browser cache -> Hard refresh (Ctrl+Shift+R)
“500 Internal Server Error”	Check Apache error log for PHP fatal error
“reCAPTCHA not working”	Using production keys on localhost -> Use test keys
“File upload fails”	upload_max_filesize too low -> Increase in php.ini

Where to Find Logs

```
# XAMPP Windows
C:\xampp\apache\logs\error.log
C:\xampp\php\logs\php_error_log

# XAMPP macOS/Linux
/Applications/XAMPP/xamppfiles/logs/error.log # macOS
/opt/lampp/logs/error.log # Linux

# Laragon
C:\laragon\var\log\apache2\error.log

# Docker
docker-compose logs -f app

# Project logs
/var/www/vueports/logs/itn_raw.txt
```

11. VS Code Setup

11.1 Recommended Extensions

- PHP Intelephense (code intelligence)
- MySQL (database management)
- Apache Conf (syntax highlighting)
- Prettier (code formatting)
- ESLint (JavaScript linting)
- Live Server (optional, for static preview)

11.2 Launch Configuration

Create `.vscode/launch.json`:

```
{
  "version": "0.2.0",
  "configurations": [
    {
      "name": "Listen for Xdebug",
      "type": "php",
      "request": "launch",
      "port": 9003,
      "pathMappings": {
        "/var/www/html": "${workspaceFolder}"
      }
    }
  ]
}
```

11.3 Xdebug Setup (Optional)

```
; Add to php.ini
zend_extension=xdebug
xdebug.mode=debug
xdebug.start_with_request=yes
xdebug.client_port=9003
xdebug.client_host=localhost
```

12. Testing PayFast Locally

12.1 Sandbox Mode

```
# .env must have:
PAYFAST_SANDBOX=true
PAYFAST_MERCHANT_ID=10000100
PAYFAST_MERCHANT_KEY=46f0cd694581a
```

12.2 Test Card / Account

PayFast sandbox uses simulated payments — no real card needed.

12.3 ITN Testing Locally

PayFast ITN requires a public URL. For local testing:

Option A: ngrok (Recommended)

```
# Install ngrok: https://ngrok.com/download
ngrok http 80

# You'll get a public URL like https://abc123.ngrok.io
# Update .env:
# APP_URL=https://abc123.ngrok.io
# PAYFAST_NOTIFY_URL=https://abc123.ngrok.io/webhooks/payfast-itn.php
```

Option B: LocalTunnel

```
npm install -g localtunnel
lt --port 80
# Use provided URL for ITN testing
```

Option C: Skip ITN locally

```
# After PayFast redirect, manually verify payment in admin panel
# Or use PayFast sandbox dashboard to simulate ITN
```

12.4 Test Payment Flow

1. Go to `http://localhost/pay.php?plan=starter`
2. Fill test details
3. Complete PayFast sandbox checkout
4. Check `logs/itn_raw.txt` for ITN data
5. Verify payment status in admin dashboard

Quick Reference

```
# Common URLs (XAMPP default)
http://localhost/           # Website
http://localhost/phpmyadmin # Database GUI
http://localhost/admin/login.php # Admin portal

# MySQL CLI
mysql -u vueports_user -p

# Restart services (XAMPP)
# Windows: XAMPP Control Panel -> Stop/Start
# macOS: sudo /Applications/XAMPP/xamppfiles/xampp restart
# Linux: sudo /opt/lampp/lampp restart

# View PHP info
# Create test.php with: <?php phpinfo(); ?>
# Access: http://localhost/test.php
```

Happy Coding!

For support: njabulod.hlongwane@gmail.com | +27 (68) 826-1507